

GEOS-MPM Examples Suite Report

Generated by examples/runExampleSuite

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Running and rerunning

Each example directory contains a copyable `pfw_input_*.py` and a `runProblem` wrapper. The wrapper stages an isolated case directory under `testRunDirectory`, copies the needed PFW files, runs the particle-file writer, submits GEOS, submits dependent post-processing, and copies figures/logs back to the example `output/` directory.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter/examples
./runExampleSuite all --python /usr/tce/bin/python3 --force
./runExampleSuite all --case borehole_Ghareb --python /usr/tce/bin/python3 --force
./runExampleSuite report --python /usr/tce/bin/python3
```

If output exists, `runProblem` prompts before overwriting. Use `--force` for noninteractive reruns.

Timing summary

Example	Family	GEOS job	GEOS state	GEOS wall	GEOS CPU	Post wall	Post CPU
brazilianDisk_BSpline	Brazilian disk	6044292	COMPLETED	00:02:18	00:00:00	00:00:10	00:00:00

brazilianDisk_BSpline

Brazilian disk compression using the new SinglePointBSpline particle type. The case keeps the FMPM update path and quartz damage material but uses `particleType=1` so GEOS exercises `ParticleType::SinglePointBSpline`.

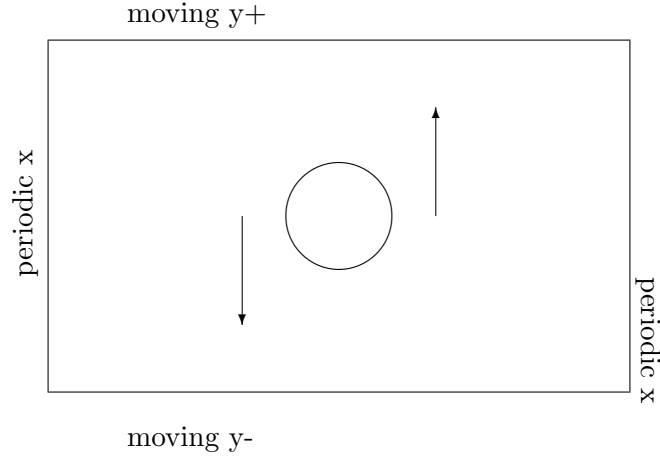


Figure 1: Schematic for brazilianDisk_BSpline.

Code features exercised.

- SinglePointBSpline particle type
- FMPM update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

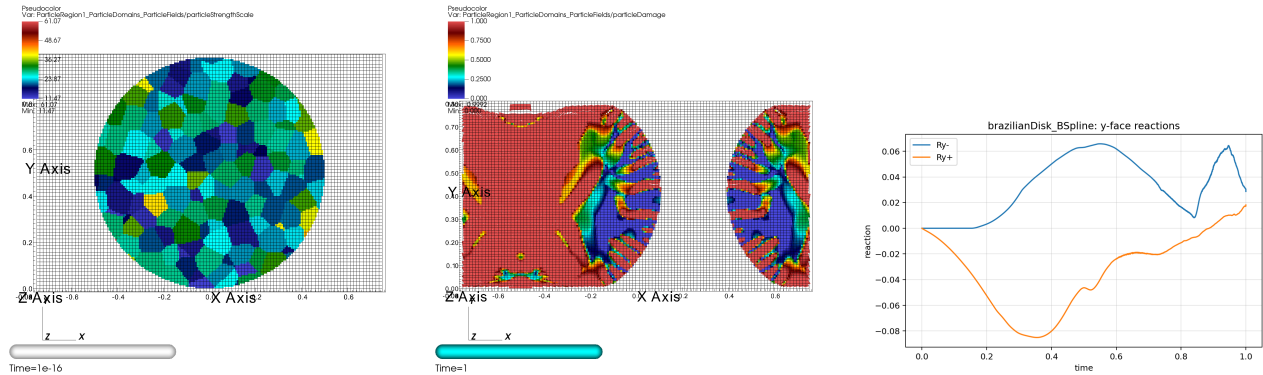


Figure 2: Initial state, final state, and reaction history for brazilianDisk_BSpline.